

Causal Semantics

Meaning as naturally equivalent update operators across independent contexts, on the stochastic-fibration compiler

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Date: August 3, 2026 **Status:** two theorems + one corollary + one exact worked example (six messages, four contexts). Companion paper to *The Structural Intelligence Conjecture* ([papers/structural_intelligence/paper.md](#)); depends on Theorem 4 of that paper (common sufficient statistic under a shared Markov screen) and instantiates the extended-program clause §5.7 as a theorem.

Abstract

Extended-program clause §5.7 of *The Structural Intelligence Conjecture* conjectures that "two symbols are equivalent when they induce naturally equivalent update operators $\Psi_{\{m, c\}}$ across independent contexts — a meaning layer that ordinary co-occurrence embeddings omit". This paper turns that clause into a theorem in the discrete-message case, and gives its natural companion instrument.

- **Theorem CS-1 (Ψ -equivalence is a congruence).** For a message space $\mathcal{M}$, a context space $\mathcal{C}$, and an update operator $\Psi : \mathcal{M} \times \mathcal{C} \rightarrow \mathcal{P}(\mathcal{X})$ sending each (m, c) to a distribution over future states, the relation

$$m_1 \sim_{\Psi} m_2 \quad :\Leftrightarrow \quad \forall c \in \mathcal{C} : \Psi(m_1, c) = \Psi(m_2, c)$$

is a proper equivalence relation on $\mathcal{M}$ (reflexive, symmetric, transitive), and it is preserved under context substitution: if $m_1 \sim_{\Psi} m_2$ then for every finite context sequence $c_1, \dots, c_n \in \mathcal{C}$, $\Psi(m_1, c_i) = \Psi(m_2, c_i)$ for every i .

- **Theorem CS-2 (Meaning quotient is the coarsest sufficient statistic on messages).** The quotient $\mathcal{M} / \sim_{\Psi}$ is the coarsest partition of $\mathcal{M}$ that preserves every downstream distribution $\Psi(\cdot, c)$. Formally, $\mathcal{M} / \sim_{\Psi}$ is a **common sufficient statistic** for the family $\{\Psi(\cdot, c) : c \in \mathcal{C}\}$ on $\mathcal{M}$ in the sense of Theorem 4 of the parent paper, and no strictly coarser partition of $\mathcal{M}$ has the same property.
- **Corollary (Causal semantics $\perp$ ambient co-occurrence).** Naïve co-occurrence embeddings group messages by the ambient distribution of the messages themselves (which tokens appear next to which), not by their downstream causal effect. Ψ -equivalence and co-occurrence-equivalence are, in general, incomparable partitions of $\mathcal{M}$. Concretely, the worked example of §4 exhibits six messages with a four-class Ψ -quotient $\{\{m_0, m_1\}, \{m_2, m_3\}, \{m_4\}, \{m_5\}\}$ and a two-class co-occurrence partition $\{\{m_0, m_2, m_4\}, \{m_1, m_3, m_5\}\}$: neither refines the other, and no single-cluster overlap coincides.

We close with an exact worked example ([experiments/causal_semantics_pair](#)) of six messages and four contexts on a four-state future space $\mathcal{X} = \{x_0, x_1, x_2, x_3\}$. Ψ is hand-built so that $m_0 \sim_{\Psi} m_1$ (identical downstream distributions on every context), $m_2 \sim_{\Psi} m_3$ (identical downstream, different from the m_0 class), and m_4, m_5 each stand alone. The distractor co-occurrence matrix is designed so that a naïve co-occurrence clustering pools messages by their ambient token signature — grouping the even-indexed messages against the odd-indexed messages — an orthogonal structure to $\sim_{\Psi}$. Every gate of the pre-registered set passes exactly.

The paper is a *reduction*, not a new representation-learning recipe. Its content is *organising*: it lets the vague §5.7 clause be read as a concrete equivalence-relation theorem, and it identifies exactly which axis of a candidate embedding is doing the load-bearing work of *meaning* (constancy of $\Psi(\cdot, c)$) versus *co-occurrence* (constancy of ambient marginals).

1. Setup

We inherit the master object of *The Structural Intelligence Conjecture* §1: a stochastic fibration $(\mathcal{Q} : \mathcal{X} \rightarrow \mathcal{Z}, \mathcal{K} : \mathcal{Z} \rightsquigarrow \mathcal{X})$ on a standard Borel $\mathcal{X}$. Fix

- a **message space** $\mathcal{M}$ (a countable or standard-Borel set of discrete messages an agent may emit or receive),
- a **context space** $\mathcal{C}$ (the same, but for the ambient state in which a message is encountered), and
- a **future-state space** $\mathcal{D}$ on which a probability measure lives, together with an **update operator**

$$\Psi : \mathcal{M} \times \mathcal{C} \rightarrow \mathcal{P}(\mathcal{D}), \quad \Psi(m, c) \in \mathcal{P}(\mathcal{D}),$$

sending each $(\text{message}, \text{context})$ pair to a distribution over future states. Operationally $\Psi(m, c)$ is the law of the next fine state when the agent is in context c and receives message m — the literal *update* the message induces on the agent's fine-state distribution.

Ψ -equivalence of messages. Two messages $m_1, m_2 \in \mathcal{M}$ are **Ψ -equivalent** iff

$$\forall c \in \mathcal{C} : \Psi(m_1, c) = \Psi(m_2, c),$$

written $m_1 \sim_{\Psi} m_2$. Two Ψ -equivalent messages induce the same downstream distribution in every possible context, and are therefore downstream-indistinguishable.

Meaning quotient. The set of equivalence classes $\mathcal{M} / \sim_{\Psi}$ is called the **meaning quotient** of $\mathcal{M}$ with respect to Ψ . Its cells are the coarsest downstream-preserving grouping of messages: two messages are pooled iff no context can tell them apart by the future they induce.

Distractor: naïve co-occurrence. For contrast, fix a **co-occurrence signature map** $\kappa : \mathcal{M} \rightarrow \mathbb{R}^{d_{\kappa}}$ (e.g. a row of an ambient message-by-token count matrix, normalised to a probability vector, or a Word2Vec-style embedding). Define

$$m_1 \sim_{\kappa} m_2 \quad :\Leftrightarrow \quad \kappa(m_1) = \kappa(m_2)$$

as **co-occurrence-equivalence** of messages. The corresponding partition $\mathcal{M} / \sim_{\kappa}$ is what a naïve embedding recovers when messages have no downstream distribution to be measured against — it groups by *which other tokens the message appears near*, not by *what the message causes*. Concrete embeddings (Word2Vec, GloVe, BERT masked-token, etc.) typically produce a continuous version whose partition is a nearest-neighbour clustering of κ ; here we take the exact discrete case where the signature $\kappa(m)$ is a finite-dimensional vector and messages with identical vectors form a co-occurrence class.

$\sim_{\Psi}$ and $\sim_{\kappa}$ are, in general, distinct equivalence relations on $\mathcal{M}$. §3 proves this in the sharp form: neither refines the other, and neither is implied by the other — the two axes of representation are formally orthogonal.

2. Theorem CS-1: Ψ -equivalence is a congruence

Setup (CS-1). Fix $\Psi : \mathcal{M} \times \mathcal{C} \rightarrow \mathcal{P}(\mathcal{D})$. Define $\sim_{\Psi}$ as in §1.

Theorem CS-1 (Ψ -equivalence is a congruence). $\sim_{\Psi}$ is a proper equivalence relation on $\mathcal{M}$: reflexive, symmetric, and transitive. Furthermore, $\sim_{\Psi}$ is preserved under context substitution: if $m_1 \sim_{\Psi} m_2$ then for every finite sequence of contexts $c_1, \dots, c_n \in \mathcal{C}$, $\Psi(m_1, c_i) = \Psi(m_2, c_i)$ for every $i \in \{1, \dots, n\}$.

Proof.

(Reflexivity.) For every $m \in \mathcal{M}$ and every $c \in \mathcal{C}$, $\Psi(m, c) = \Psi(m, c)$ by definition of a function. Hence $m \sim_{\Psi} m$ for every m .

(Symmetry.) Suppose $m_1 \sim_{\Psi} m_2$. Then for every $c \in \mathcal{C}$, $\Psi(m_1, c) = \Psi(m_2, c)$, so $\Psi(m_2, c) = \Psi(m_1, c)$ by symmetry of equality. Hence $m_2 \sim_{\Psi} m_1$.

(Transitivity.) Suppose $m_1 \sim_{\Psi} m_2$ and $m_2 \sim_{\Psi} m_3$. Then for every $c \in \mathcal{C}$, $\Psi(m_1, c) = \Psi(m_2, c) = \Psi(m_3, c)$ by transitivity of equality applied pointwise. Hence $m_1 \sim_{\Psi} m_3$.

(Context substitution.) Suppose $m_1 \sim_{\Psi} m_2$ and let $c_1, \dots, c_n \in \mathcal{C}$ be any finite context sequence. By definition of $\sim_{\Psi}$, $\Psi(m_1, c) = \Psi(m_2, c)$ for **every** $c \in \mathcal{C}$; in particular this holds at each c_i . $\square$

Remark (why congruence). In the algebraic sense, a congruence on a set with a family of unary operations $\{f_c\}_{c \in \mathcal{C}}$ is an equivalence relation preserved by every f_c . Setting $f_c(m) := \Psi(m, c) \in \mathcal{P}(\mathcal{D})$, the statement $m_1 \sim_{\Psi} m_2 \Rightarrow f_c(m_1) = f_c(m_2)$ for every c is *exactly* congruence with respect to the family $\{f_c\}_{c \in \mathcal{C}}$. So $\sim_{\Psi}$ is the largest congruence on $\mathcal{M}$ compatible with the update-operator family. This is a recasting of the well-known fact that the finest partition respecting a family of maps is the intersection of the level-set partitions of each map — but interpreted here as the *meaning* of a symbol reduces to its downstream causal role under every context.

Consequence. Two Ψ -equivalent messages can be substituted for one another in any context without altering the induced downstream distribution. This is the operational meaning of "same meaning" in this framework: substitutability preserves observable effect.

3. Theorem CS-2: Meaning quotient is the coarsest sufficient statistic on messages

Setup (CS-2). Fix $\Psi : \mathcal{M} \times \mathcal{C} \rightarrow \mathcal{P}(\mathcal{D})$. For each $c \in \mathcal{C}$, $\Psi(\cdot, c) : \mathcal{M} \rightarrow \mathcal{P}(\mathcal{D})$ is a function on $\mathcal{M}$. Read this as a family of $\mathcal{M}$ -indexed distributions

$$\{ \Psi(\cdot, c) : c \in \mathcal{C} \}$$

in the sense of Theorem 4 of the parent paper — a family of tasks whose target is a distribution over $\mathcal{D}$ and whose input is a message $m \in \mathcal{M}$.

Theorem CS-2 (Meaning quotient is the coarsest sufficient statistic on messages). *The quotient $\pi_{\Psi} : \mathcal{M} \rightarrow \mathcal{M} / \sim_{\Psi}$ is the coarsest partition of $\mathcal{M}$ that preserves every downstream distribution in the family $\{\Psi(\cdot, c) : c \in \mathcal{C}\}$. Formally, π_{Ψ} is a common sufficient statistic (in the Theorem-4 sense) for the family on $\mathcal{M}$, and no strictly coarser partition of $\mathcal{M}$ is common sufficient for the family.*

Proof.

(Sufficiency.) Let $c \in \mathcal{C}$ and take two messages $m_1, m_2 \in \mathcal{M}$ lying in the same Ψ -class, i.e. $\pi_{\Psi}(m_1) = \pi_{\Psi}(m_2)$. By definition of $\sim_{\Psi}$, $\Psi(m_1, c) = \Psi(m_2, c)$. So the value of $\Psi(\cdot, c)$ is constant on each Ψ -class — equivalently, $\Psi(\cdot, c)$ factors through π_{Ψ} :

$$\Psi(m, c) = \hat{\Psi}(\pi_{\Psi}(m), c)$$

for a well-defined $\hat{\Psi} : (\mathcal{M} / \sim_{\Psi}) \times \mathcal{C} \rightarrow \mathcal{P}(\mathcal{D})$. This is the factorisation $Y_{\alpha} \perp\!\!\!\perp \mathcal{M} \mid \pi_{\Psi}(\mathcal{M})$ in the Theorem-4 template with $Y_{\alpha} := \Psi(\cdot, c) \in \mathcal{P}(\mathcal{D})$ and $X := \mathcal{M}$. Hence π_{Ψ} is a common sufficient statistic for the family.

(Coarsest.) Suppose $\pi' : \mathcal{M} \rightarrow \mathcal{S}'$ is a strictly coarser partition of $\mathcal{M}$ than π_{Ψ} — i.e. π' identifies at least one pair of messages that π_{Ψ} distinguishes. Take such a pair m_1, m_2 with $\pi'(m_1) = \pi'(m_2)$ but $\pi_{\Psi}(m_1) \neq \pi_{\Psi}(m_2)$. Then $m_1 \not\sim_{\Psi} m_2$, so there exists a context $c^* \in \mathcal{C}$ such that $\Psi(m_1, c^*) \neq \Psi(m_2, c^*)$. If π' were common sufficient for the family, then $\Psi(\cdot, c^*)$ would factor through π' and take the same value on m_1 and m_2 (since $\pi'(m_1) = \pi'(m_2)$) — contradiction. So no strictly coarser partition is common sufficient, and π_{Ψ} is the coarsest common sufficient statistic on $\mathcal{M}$. $\square$

Corollary (Causal semantics $\perp$ ambient co-occurrence). *The meaning quotient $\mathcal{M} / \sim_{\Psi}$ and the co-occurrence quotient $\mathcal{M} / \sim_{\kappa}$ are, in general, distinct partitions of $\mathcal{M}$. In particular, $\sim_{\Psi}$ need not refine $\sim_{\kappa}$, and $\sim_{\kappa}$ need not refine $\sim_{\Psi}$ — the two axes are incomparable. This is because $\sim_{\Psi}$ is determined by the map $m \mapsto \Psi(m, \cdot) \in \mathcal{P}(\mathcal{D})^{\mathcal{C}}$, while $\sim_{\kappa}$ is determined by the map $m \mapsto \kappa(m) \in \mathbb{R}^{\mathcal{D}}$; there is no formal constraint relating these two maps, and the worked example of §4 exhibits a case where the partitions are strictly incomparable.*

Proof. Take the six-message, four-context worked example of §4. The Ψ -quotient there is

$$\mathcal{M} / \sim_{\Psi} = \{ \{m_0, m_1\}, \{m_2, m_3\}, \{m_4\}, \{m_5\} \} \quad (4 \text{ classes})$$

and the co-occurrence quotient is

$$\mathcal{M} / \sim_{\kappa} = \{ \{m_0, m_2, m_4\}, \{m_1, m_3, m_5\} \} \quad (2 \text{ classes}).$$

No cell of one partition is a union of cells of the other: $\{m_0, m_1\}$ is neither a subset nor a superset of $\{m_0, m_2, m_4\}$ (they share only m_0 but each contains a message the other does not). So neither partition refines the other. $\square$

Consequence (operational). A representation that groups messages by ambient co-occurrence (Word2Vec, GloVe, contextualised token embeddings without downstream-consequence signal, etc.) will in general *fail* to recover the meaning quotient of §1. What it recovers is a different partition — the one adapted to which tokens co-occur with which, not to which futures each token causes. The two axes may agree in some corpora (where downstream causal role correlates with ambient co-occurrence, e.g. in synonym pairs that also co-occur with the same words), but they are formally independent. Where they diverge, a naïve embedding grouped by $\sim_{\kappa}$ is not learning meaning in the sense of §1.

4. Worked example: six messages, four contexts

Message space. $\mathcal{M} = \{m_0, m_1, m_2, m_3, m_4, m_5\}$ (six discrete symbols, no additional structure).

Context space. $\mathcal{C} = \{c_0, c_1, c_2, c_3\}$ (four discrete contexts).

Future-state space. $\mathcal{X} = \{x_0, x_1, x_2, x_3\}$ (four discrete future states).

Update operator Ψ . Hand-built so that four distinct downstream-distribution *classes* arise:

$$\begin{aligned} \Psi(m_0, \cdot) &= \Psi(m_1, \cdot) = D_A(\cdot) && [\text{class A}] \\ \Psi(m_2, \cdot) &= \Psi(m_3, \cdot) = D_B(\cdot) && [\text{class B}] \\ \Psi(m_4, \cdot) &= D_C(\cdot) && [\text{class C}] \\ \Psi(m_5, \cdot) &= D_D(\cdot) && [\text{class D}] \end{aligned}$$

where the four class-conditional context maps are (rows = contexts, columns = future states x_0, x_1, x_2, x_3):

context	$D_A(c)$	$D_B(c)$	$D_C(c)$	$D_D(c)$
c_0	(0.50, 0.50, 0.00, 0.00)	(0.00, 0.00, 0.50, 0.50)	(1.00, 0.00, 0.00, 0.00)	(0.00, 0.00, 0.00, 1.00)
c_1	(0.25, 0.25, 0.25, 0.25)	(0.70, 0.10, 0.10, 0.10)	(0.10, 0.70, 0.10, 0.10)	(0.10, 0.10, 0.70, 0.10)
c_2	(0.30, 0.30, 0.20, 0.20)	(0.20, 0.20, 0.30, 0.30)	(0.50, 0.50, 0.00, 0.00)	(0.00, 0.00, 0.50, 0.50)
c_3	(0.40, 0.10, 0.40, 0.10)	(0.10, 0.40, 0.10, 0.40)	(0.25, 0.25, 0.25, 0.25)	(0.10, 0.40, 0.40, 0.10)

Every row is a valid probability vector (sums to 1). By direct inspection at c_0 , D_A , D_B , D_C , D_D are four distinct distributions on $\mathcal{X}$; so the Ψ -classes are pairwise distinct. By construction, within each class the downstream distribution is identical in every context, so the Ψ -quotient is exactly

$$\mathcal{M} / \sim_{\Psi} = \{ \{m_0, m_1\}, \{m_2, m_3\}, \{m_4\}, \{m_5\} \}.$$

Distractor: co-occurrence signature. Hand-built ambient co-occurrence matrix $\kappa : \mathcal{M} \rightarrow \mathbb{Z}^4_{\geq 0}$ (rows = messages, columns = ambient tokens t_0, t_1, t_2, t_3):

message	$\kappa(m)$
m_0	(4, 4, 1, 1)
m_1	(1, 1, 4, 4)
m_2	(4, 4, 1, 1)

m ₃	(1, 1, 4, 4)
m ₄	(4, 4, 1, 1)
m ₅	(1, 1, 4, 4)

Even-indexed messages $\{m_0, m_2, m_4\}$ share the signature $(4, 4, 1, 1)$; odd-indexed messages $\{m_1, m_3, m_5\}$ share the signature $(1, 1, 4, 4)$. Under $\sim_K$ the co-occurrence quotient is therefore

$$\mathcal{M} / \sim_K = \{ \{m_0, m_2, m_4\}, \{m_1, m_3, m_5\} \}.$$

This is *orthogonal* to the Ψ -quotient: m_0 and m_1 are $\sim_\Psi$ but not $\sim_K$; m_0 and m_2 are $\sim_K$ but not $\sim_\Psi$. Neither partition refines the other, and the Corollary of §3 is instantiated exactly.

Instrument gates. The witness (`experiments/causal_semantics_pair`) computes both partitions exactly (no random seeds, no Monte Carlo) and verifies:

1. `cs1_psi_equivalence_is_reflexive_symmetric_transitive`: the binary relation $\sim_\Psi$ on $\mathcal{M} \times \mathcal{M}$ is verified reflexive, symmetric, and transitive by exhaustive enumeration of all $36 = 6 \times 6$ ordered pairs and, for transitivity, all $216 = 6^3$ ordered triples.
2. `cs2_psi_quotient_has_four_classes`: the equivalence classes of $\sim_\Psi$ on $\mathcal{M}$ are exactly $\{m_0, m_1\}, \{m_2, m_3\}, \{m_4\}, \{m_5\}$.
3. `cs2_psi_quotient_is_common_sufficient`: for every context $c \in \mathcal{C}$, the downstream distribution $\Psi(\cdot, c)$ is constant within each Ψ -class (i.e., π_Ψ is a common sufficient statistic for the family $\{\Psi(\cdot, c) : c \in \mathcal{C}\}$ on $\mathcal{M}$).
4. `cs_cooccurrence_partition_differs_from_psi_quotient`: the co-occurrence partition $\mathcal{M} / \sim_K$ and the Ψ -quotient $\mathcal{M} / \sim_\Psi$ are distinct partitions of $\mathcal{M}$ (as sets of cells). Neither refines the other, and no cell of either partition coincides with a cell of the other.

All four gates pass exactly.

5. Relation to the SIC framework

Theorem CS-1 promotes the extended-program §5.7 clause of the parent paper from a *research direction* ("two symbols are equivalent when they induce naturally equivalent update operators $\Psi_{\{m, c\}}$ across independent contexts — a meaning layer that ordinary co-occurrence embeddings omit") to a *theorem* in the discrete-message case: $\sim_\Psi$ is a proper equivalence relation, closed under context substitution. Theorem CS-2 identifies the meaning quotient $\mathcal{M} / \sim_\Psi$ as the exact object promised by §5.7 — the coarsest common sufficient statistic on messages for the context-conditional downstream family — thereby applying Theorem 4 of the parent paper one level up, at the message layer rather than the world-state layer.

The corollary makes the orthogonality claim precise: co-occurrence embeddings and causal-semantic embeddings are, in general, independent partitions of $\mathcal{M}$, and the worked example exhibits a concrete case where they are strictly incomparable. This is the *meaning layer that ordinary co-occurrence embeddings omit* of §5.7, made into a theorem and a numerical witness.

Together with

- Theorems CG-1, CG-2 (Fisher geometry and holonomy of concern from *Concern as Fiber Geometry*),
- Theorems CT-1, CT-2 (MDL identifiability and Boltzmann ecology from *Compiler Tomography*),
- Theorem SA-1 (antecedent taxonomy from *Sufficient Antecedents for Cross-Task Stability*),
- Theorems AF-1, AF-2 (abstraction frontier as a Pareto antichain from *The Abstraction Frontier*), and
- Theorems AG-1, AG-2 (viability under bounded leakage from *Alignment as Ensemble Governance*),

Theorems CS-1, CS-2 give the SIC extended program another explicit theorem-instrument pair. The remaining §5 constructs (theory atlas §5.5, representation-repair calculus §5.8, autocatalytic artwork §5.10) remain open — each is a

direction rather than a theorem.

The CS framing sharpens the SIC honesty condition on representation choice: an embedding that reports "these two symbols mean the same thing" must be checkable against the meaning quotient $\mathcal{M} / \sim_{\Psi}$, not merely against a co-occurrence quotient $\mathcal{M} / \sim_{\kappa}$. Where the two agree, no distinction is drawn; where they disagree, the embedding is naming co-occurrence and calling it meaning.

6. Limitations

- **Discrete messages, discrete contexts.** CS-1 and CS-2 are stated and proved in the countable-message, countable-context case. The statements extend directly to standard-Borel $\mathcal{M}$ and $\mathcal{C}$ under the usual measure-theoretic definitions of equivalence (a.s. equality of the two update laws for every context), but the worked-example instrument uses the discrete case for exact witness.
 - **Exact operator equality.** $\sim_{\Psi}$ requires exact equality of the downstream distributions across every context. In practice, one measures a metric on $\mathcal{P}(\mathcal{D})$ (total variation, KL, Wasserstein) and works with an ε -relaxation $\sim_{\Psi}^{\varepsilon}$. The proofs of CS-1 and CS-2 go through for the ε -relaxation with the obvious modifications (reflexivity holds for $\varepsilon \geq 0$, transitivity holds up to 2ε in the triangle-inequality sense for any metric, sufficiency holds up to ε), at the cost of a quantitative rather than exact statement.
 - **No estimator.** This paper does not construct an estimator of $\sim_{\Psi}$ from data; it exhibits the object as an equivalence relation and a coarsest sufficient statistic. A follow-up instrument would study sample-efficient recovery of $\sim_{\Psi}$ from finite context / outcome samples — the natural bridge to the SIC-C-c learnability question of the parent paper §2.5.
 - **Co-occurrence as a straw-man.** The distractor here is intentionally the *simplest* co-occurrence signature (exact row equality of an ambient-token count matrix). Modern embeddings (contextualised transformers, joint text-and-behaviour models) will already partially recover downstream-effect structure to the extent their training loss couples message identity to a behaviourally-relevant signal. The orthogonality claim of §3 is formally correct at the level of *what the embedding is a function of*, but the practical distance between $\sim_{\Psi}$ and $\sim_{\kappa}$ for a specific embedding is an empirical question.
 - **Not a Lean formalisation.** The proofs of CS-1 and CS-2 are elementary set-theoretic arguments (proper equivalence relation, finest partition respecting a family of maps). Formalising them in Lean 4 alongside Theorem 4's core in `formal/structural-intelligence/StructuralIntelligence/` would give the meaning quotient a machine-checked object; that work is not yet done.
 - **Does not solve meaning.** CS-1 is a *reduction*: it says the right object of *meaning* under this framework is the Ψ -quotient, and the right check against a candidate embedding is the coarsest-common-sufficient-statistic property. It says nothing about *how to construct* an embedding whose partition equals $\sim_{\Psi}$ — that is where the actual work of semantic representation learning lies.
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7. Reproduction

```
python3 experiments/causal_semantics_pair/experiment.py
```

Full development is in the parent paper's §5.7 and the master notes file `notes/structural_intelligence_conjecture.md`.